National VPN Vault

Secure Network Communication between Riyadh and Neom

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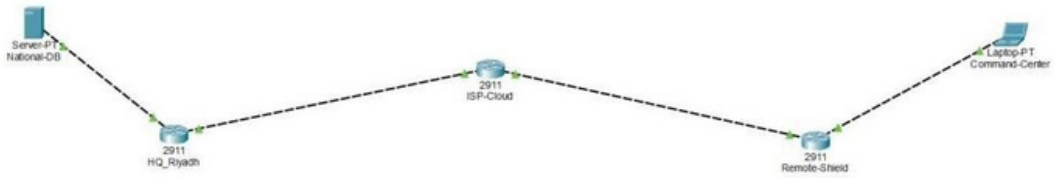
National-VPN Vault

This project focuses on designing a secure network using VPN technology to enable safe and encrypted communication between two remote sites.

The implementation uses IPsec as the main security framework, along with IKE/ISAKMP for key exchange, AES for encryption, SHA for integrity, and Diffie-Hellman Group 2 for secure key generation.

The configuration process included setting up routers, assigning IP addresses, configuring routing, applying VPN security settings, and finally testing the connection to ensure that the secure tunnel was successfully established.

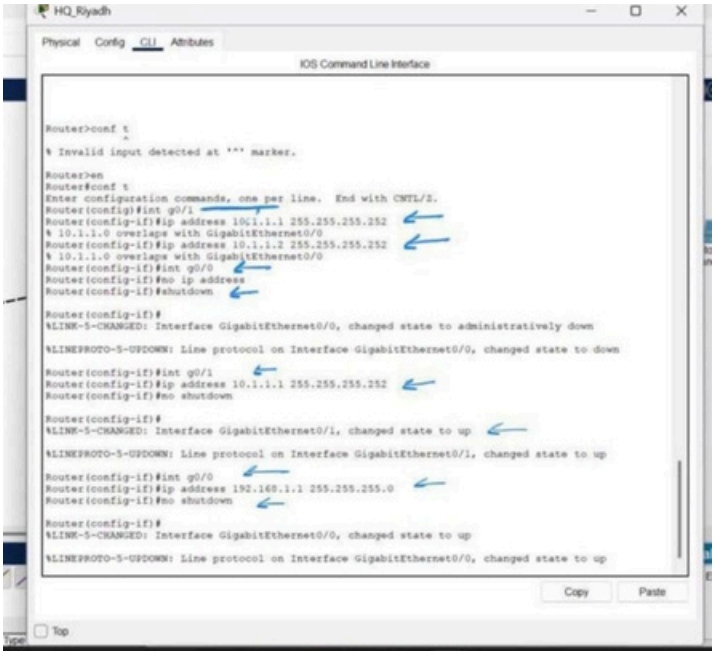
1. Network Design



Device	Interface	IP Address	Subnet Mask
National-DB	Fa0	192.168.1.10	255.255.255.0
HQ_Riyadh	Gig0/0	192.168.1.1	255.255.255.0
HQ_Riyadh	Gig0/1	10.1.1.1	255.255.255.252
ISP-Cloud	Gig0/0	10.1.1.2	255.255.255.252
ISP-Cloud	Gig0/1	10.2.2.1	255.255.255.252
Remote-Shield	Gig0/0	10.2.2.2	255.255.255.252
Remote-Shield	Gig0/1	192.168.2.1	255.255.255.0
Laptop-PT Command-Center	Fa0	192.168.2.10	255.255.255.0

This WAN connects HQ_Riyadh and Remote-Shield through an ISP using a Site-to-Site VPN over a WAN infrastructure. A Site-to-Site VPN is used to create a secure encrypted tunnel over the internet for safe communication between both sites and the National-DB server.

2. Configuration Steps



The configuration process followed a systematic approach. First, physical interfaces were activated and assigned unique IP addresses. Static routing was established using default routes (0.0.0.0 0.0.0.0) on the HQ and Branch routers, directing all outbound traffic toward the ISP to ensure basic end-to-end connectivity before applying security layers



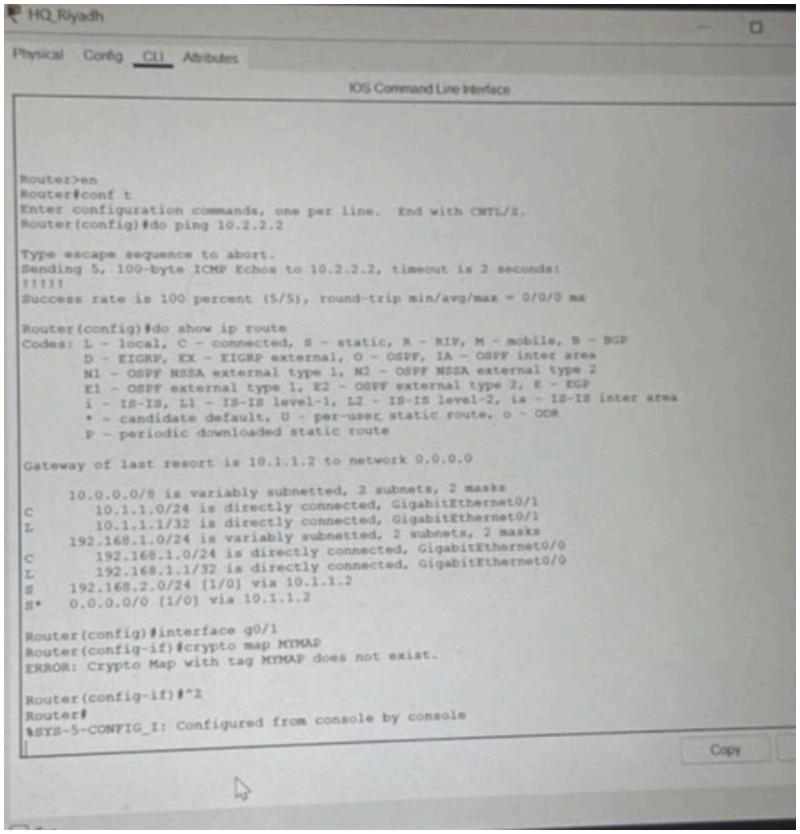
This screenshot illustrates the initial configuration of the Neom (Remote-Shield) router. It shows the assignment of IP addresses to the interfaces: GigabitEthernet 0/0 (connecting to the ISP) and GigabitEthernet 0/1 (the internal LAN). Additionally, it demonstrates the configuration of a Static Default Route toward the ISP gateway (10.2.2.1) to enable external communication. The final part of the image shows a successful Ping test to the Riyadh gateway (10.1.1.1) with a success rate of verifying basic end-to-end connectivity before applying security layer ,(!!!!) 80%

2- Configuration Steps con't

The successful Ping results (!!!!!) and active encapsulation counters verify that the secure tunnel is operational and the connection is stable

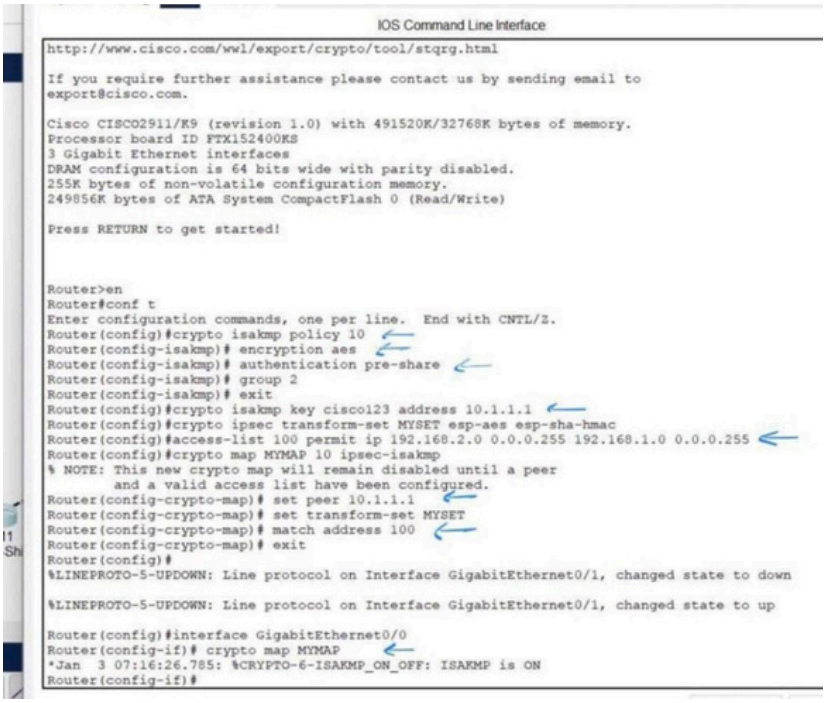


This screenshot displays a successful Ping test (!!!!!) between the routers, verifying that basic layer-3 connectivity is established and the path between Riyadh and Neom is reachable

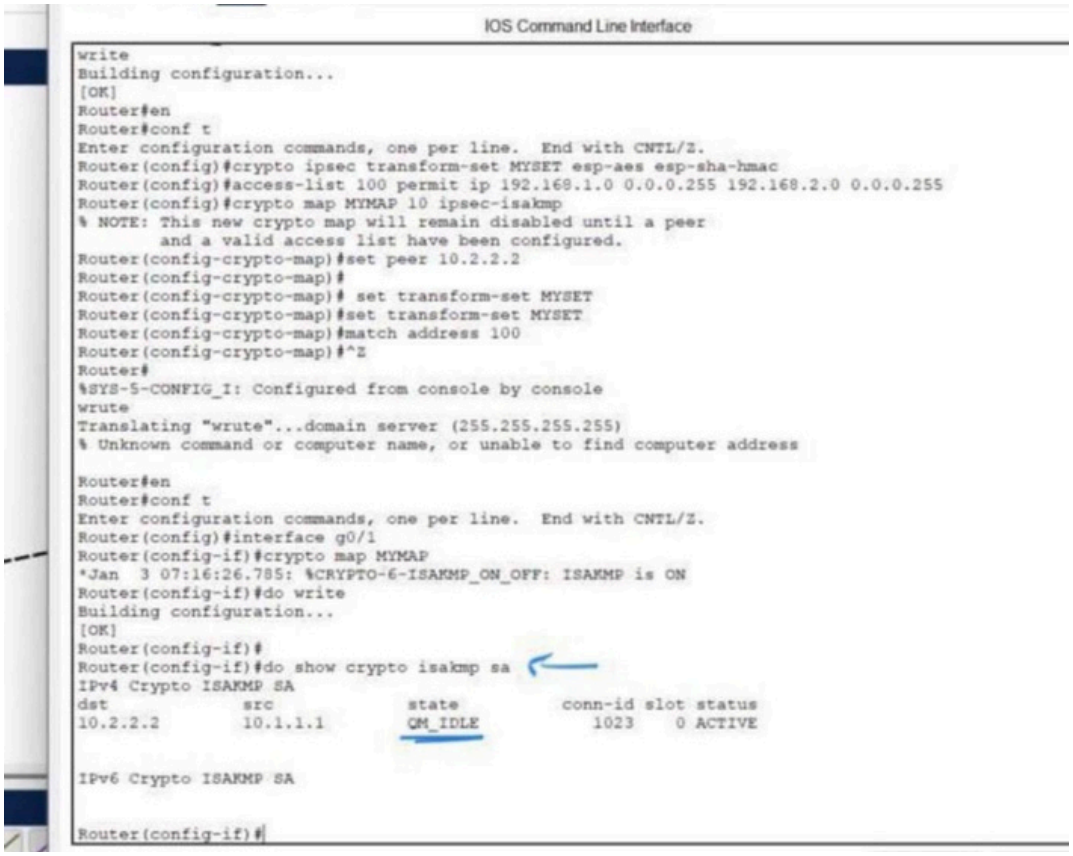


3 . Security Methods

This screenshot shows the configuration of Phase 1 and Phase 2 for a Site-to-Site IPsec VPN. It includes setting the ISAKMP policy with AES encryption, defining the pre-shared key, and applying the crypto map to the outgoing interface to secure the connection

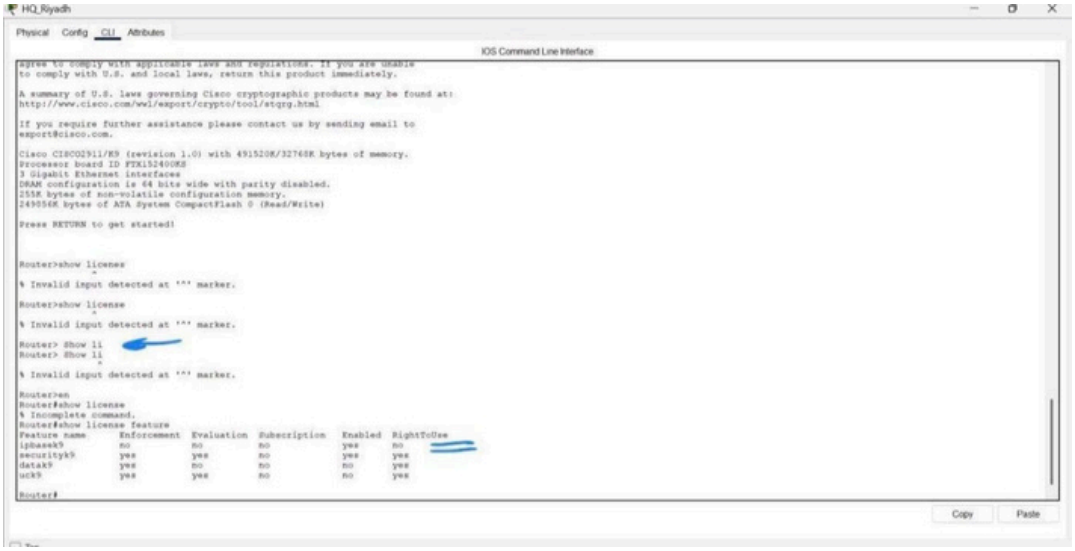


This image illustrates the verification stage using the show crypto isakmp sa command. The status QM_IDLE and ACTIVE confirm that the VPN tunnel has been successfully established and the secure .connection between the two peers is now fully operational

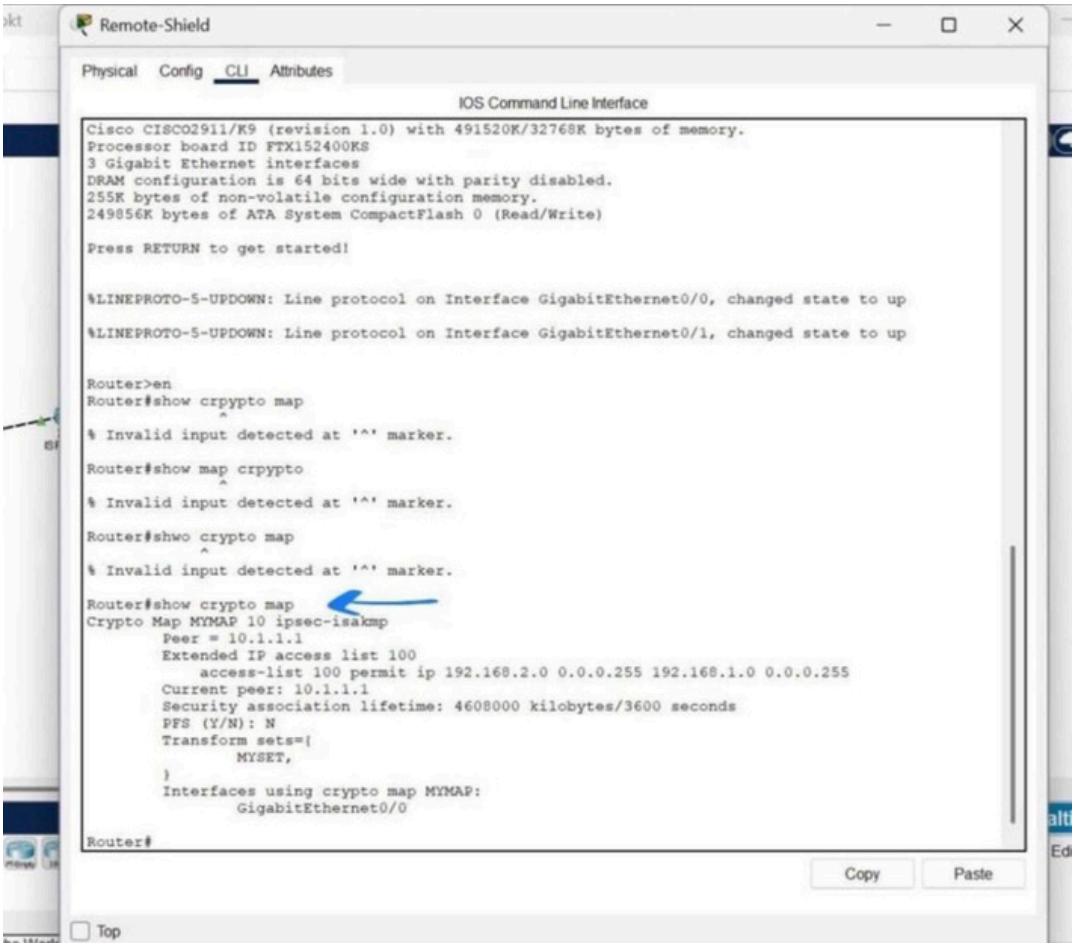


3. Security methods

This image shows the verification of router features using the show license feature command. The Right To Use status for the 'security k9' package is enabled, which is .essential for supporting IPsec VPN and cryptographic functions on Cisco routers

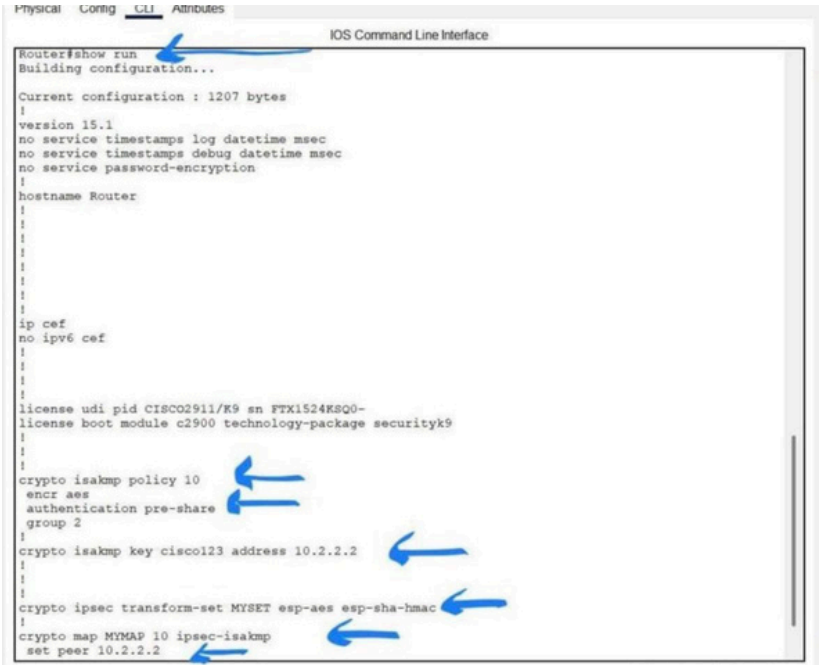


This screenshot displays the output of the show crypto map command, which verifies the configuration details of 'MYMAP'. It confirms the peer IP address (10.1.1.1), the traffic allowed by ACL 100, and ensures the crypto map is correctly bound to the GigabitEthernet0/0 interface



3. Security Methods

This screenshot displays the output of the show running-config command, providing a complete overview of the applied settings. It verifies the activation of the security technology package and lists all configured IPsec VPN parameters, including the ISAKMP policy, transform-set, and crypto map details

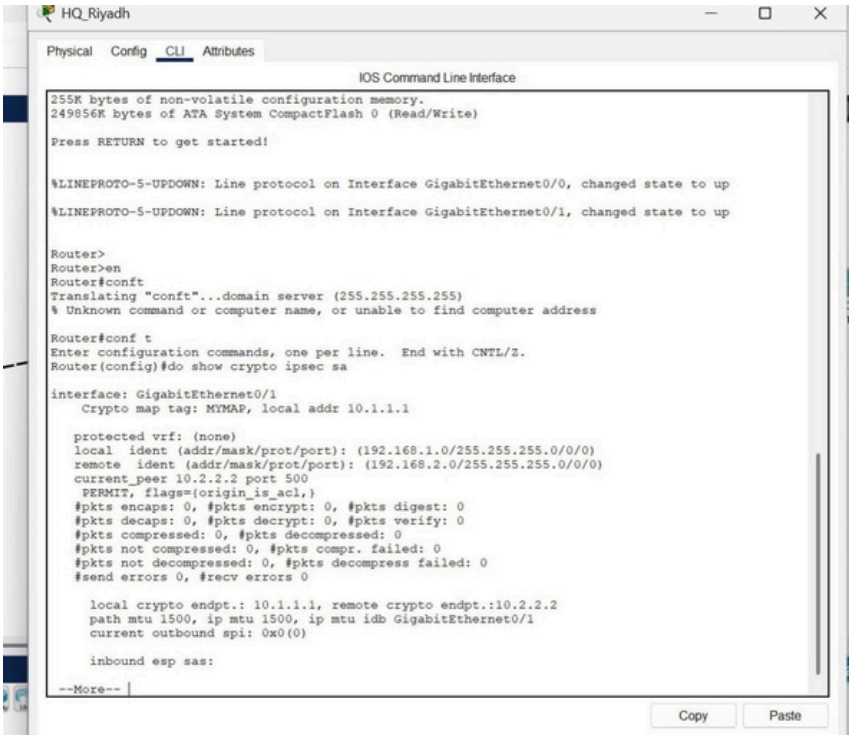


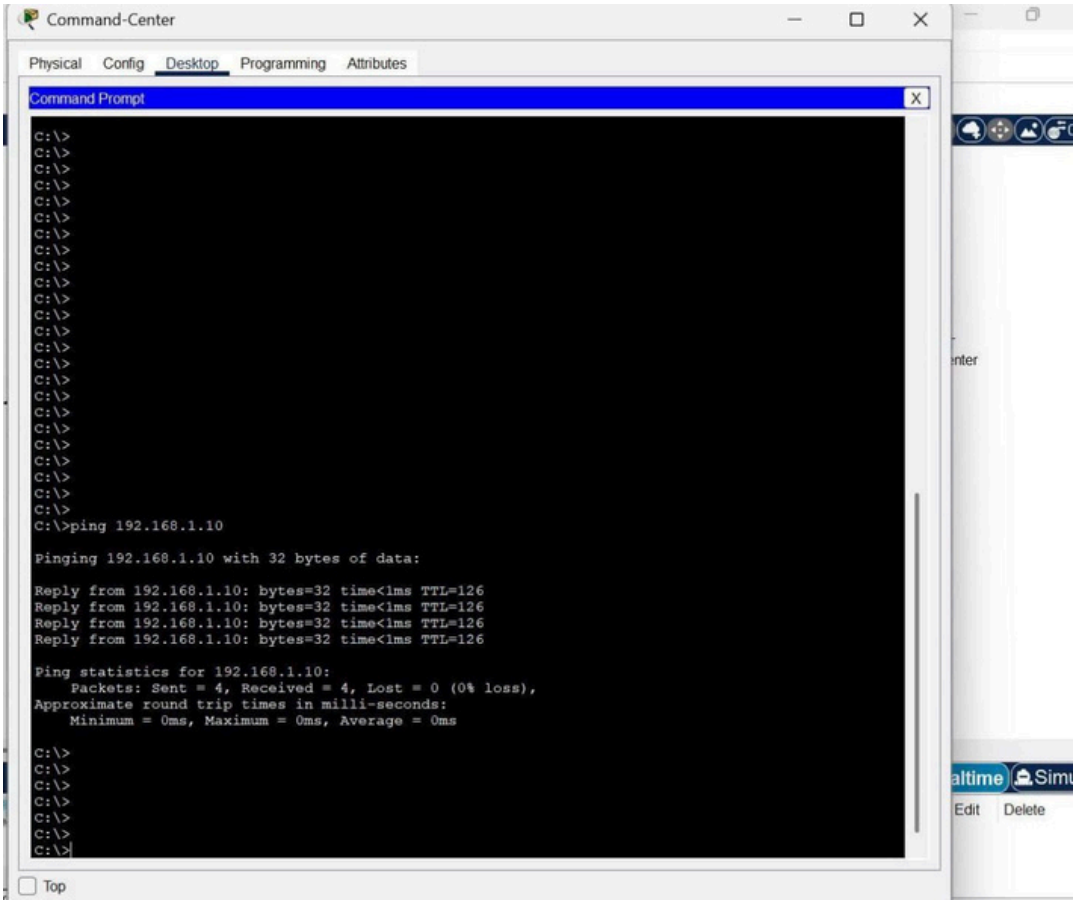
Protocols and Algorithms

Protocols/Algorithms	Purpose	Standard
IPsec	Core VPN tunneling framework	RFC 4301
IKE/ISAKMP	Phase 1 key exchange and SA negotiation	RFC 2409
AES	Symmetric encryption of data payload	NIST FIPS 197
SHA-HMAC	Data integrity / authentication	NIST FIPS 180-4
Diffie-Hellman Group 2	Secure key exchange between peers	RFC 2409
ESP	Encapsulates & encrypts packets inside tunnel	RFC 4303

4. Testing and Results

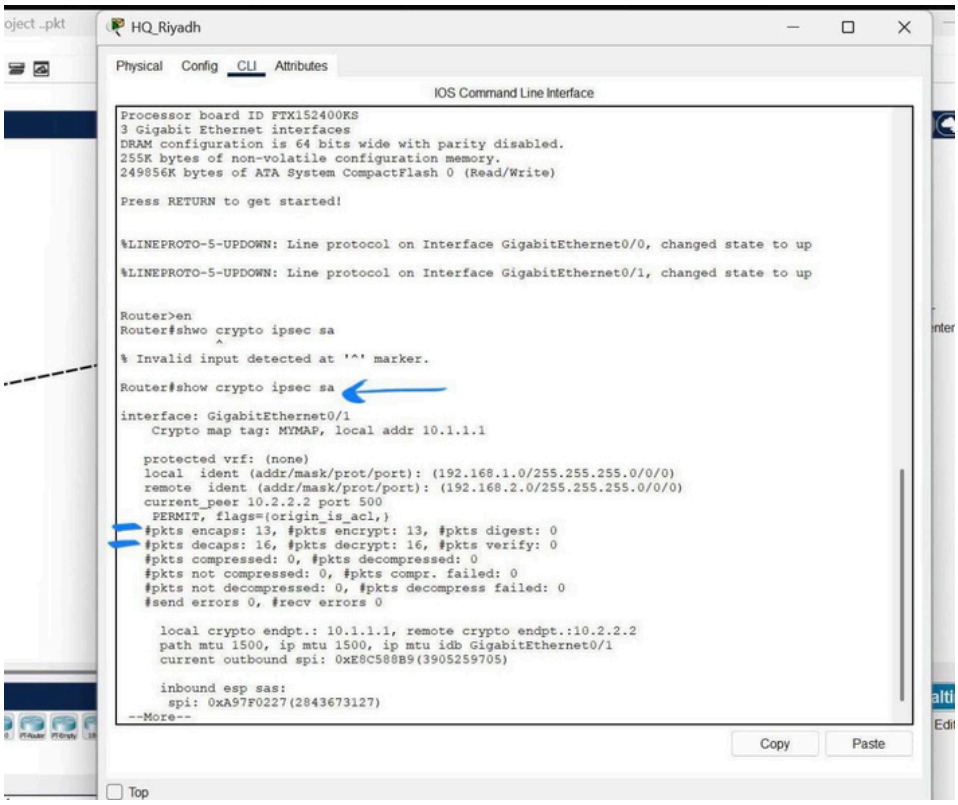
This screenshot displays the output of the show crypto ipsec sa command, which is used to verify the Phase 2 IPsec tunnel details. It shows the active security associations between the local address (10.1.1.1) and the remote peer (10.2.2.2), confirming that data traffic between the specified subnets is ready to be encrypted and encapsulated





4. Testing and Results cont...

This screenshot confirms the successful end-to-end encryption of data. The increment in #pkts encrypt and #pkts decrypt counters (showing 13 and 16 packets) provides definitive proof that the IPsec tunnel is actively protecting traffic. It demonstrates that data is being securely encapsulated and exchanged between the two sites



This screenshot showcases the end-user access via a web browser to the 'National Shield - Secure Access Portal' using the internal IP (192.168.1.10). It demonstrates the successful delivery of web services over the encrypted IPsec VPN tunnel, ensuring that sensitive data is only accessible to authorized personnel through a secure connection



6. Reflection

• Security License Activation
The most critical initial challenge was that the crypto isakmp and crypto ipsec commands were not available on the routers. This happened because the Security Technology Package was not yet active. I resolved this by running show license feature, confirming securityk9 was listed, and ensuring the license boot module c2900 technology-package securityk9 line was present in the running config after reloading.

• Interface Overlap Error
When first assigning IP addresses to GigabitEthernet0/1 on HQ_Riyadh, the router returned the error % 10.1.1.0 overlaps with GigabitEthernet0/0. This was because the same subnet had been accidentally assigned to both interfaces. I resolved it by running no ip address on the incorrect interface, then reassigning the correct /30 address.

• Crypto Map Not Found on HQ_Riyadh
When trying to apply the crypto map to the interface, the router returned ERROR: Crypto Map with tag MYMAP does not exist. This was because the crypto map had not yet been created — the commands were entered out of order. I resolved it by completing the full crypto map configuration first, then re-entering the interface and applying it.

• First Ping Packet Drop (.!!!!)
On the first ping test from Remote-Shield to HQ_Riyadh, the result was .!!!! (80% success). This is expected behavior: the first packet is dropped while IKE negotiates the tunnel. Subsequent pings and end-to-end pings after the tunnel was established all returned 100% success.

• Tunnel State: QM_IDLE / ACTIVE
Initially, show crypto isakmp sa returned no entries, indicating the tunnel had not been triggered. I resolved this by generating interesting traffic (a ping between the two LANs), which caused IKE to negotiate the Phase 1 SA. After that, the state correctly showed QM_IDLE / ACTIVE.

Overall, this project deepened my understanding of how IPsec VPNs function at the protocol level — from the IKE handshake and key exchange through to the ESP encapsulation of data packets — and how to systematically troubleshoot each phase using Cisco IOS verification commands.

7.References

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